

Quadratic Reciprocity

Unless otherwise specified, p is an ^{odd} prime.

We want to investigate solutions of the congruence equation

$$x^2 \equiv a \pmod{p}, \quad a \in \mathbb{Z}.$$

Example: when $p = 17$, $x^2 \equiv a \pmod{p}$ is solvable

when $a \equiv 0, 1, 4, 9, 16, 8, 2, 15, 13 \pmod{17}$.

Definition: For $a \in \mathbb{Z}$ s.t. $p \nmid a$,

(q.r.) a is a quadratic residue mod p if $x^2 \equiv a \pmod{p}$ is solvable;

(q.n.r.) a is a quadratic nonresidue mod p if $x^2 \equiv a \pmod{p}$ is not solvable.

Definition: (Legendre Symbol).

$$\left(\frac{a}{p}\right) = \begin{cases} +1 & \text{if } a \text{ is q.r. mod } p \\ -1 & \text{if } a \text{ is q.n.r. mod } p \\ 0 & \text{if } p \mid a. \end{cases}$$

Propⁿ: There are $\frac{p-1}{2}$ q.r. mod p , and there are $\frac{p-1}{2}$ q.n.r. mod p .

Pf: We claim that $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ are all distinct mod p .
Indeed, if $x^2 \equiv y^2 \pmod{p}$ where $x, y \in \{1, 2, \dots, \frac{p-1}{2}\}$,
then $(x+y)(x-y) \equiv 0 \pmod{p}$, so $x \equiv \pm y \pmod{p}$, which cannot happen.

Pf cont'd: Since $(p-k)^2 \equiv k^2 \pmod{p}$, the $\frac{p-1}{2}$ squares are the only q.r. mod p . Hence there are $p-1 - (\frac{p-1}{2}) = \frac{p-1}{2}$ g.n.r. mod p . □

Propⁿ: (Euler's Criterion)

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}.$$

Pf: By F.I.T., for $p \nmid a$, $a^{p-1} \equiv 1 \pmod{p}$.

$$a^{p-1} - 1 \equiv 0 \pmod{p}.$$

$$(a^{\frac{p-1}{2}} + 1)(a^{\frac{p-1}{2}} - 1) \equiv 0 \pmod{p}.$$

$$a^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}.$$

If a is q.r. mod p , then $\exists n \in \mathbb{Z}$ s.t. $a \equiv n^2 \pmod{p}$.

$$\begin{aligned} \text{then } a^{\frac{p-1}{2}} &\equiv (n^2)^{\frac{p-1}{2}} \pmod{p} \\ &\equiv n^{p-1} \pmod{p} \\ &\equiv 1 \pmod{p}. \end{aligned}$$

Consider $f(x) = x^{\frac{p-1}{2}} - 1$.

Since $\deg f = \frac{p-1}{2}$, $f(x) \equiv 0 \pmod{p}$ has at most $\frac{p-1}{2}$ solutions mod p . Since the $\frac{p-1}{2}$ q.r. mod p are all roots of this equation, the g.n.r. mod p cannot be roots of this equation.

$\therefore$ If a is g.n.r. mod p , $a^{\frac{p-1}{2}} \equiv -1 \pmod{p}$.

if $p \mid a$, $a^{\frac{p-1}{2}} \equiv 0 \pmod{p}$. Hence, $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$. □₂

Propⁿ: i) $\left(\frac{a^2}{p}\right) = 1$ for $p \nmid a$

ii) If $a \equiv b (p)$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$.

iii) $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$.

Pf: i), ii) obvious.

For iii), if $p \mid a$ or $p \mid b$, then $\left(\frac{ab}{p}\right) = 0$.

on the other hand, $ab \equiv 0 (p)$, so $\left(\frac{ab}{p}\right) = 0$.

if $p \nmid a$ & $p \nmid b$, then by Euler's Criterion,

$$\left(\frac{a}{p}\right)\left(\frac{b}{p}\right) \equiv \left(a^{\frac{p-1}{2}}\right)\left(b^{\frac{p-1}{2}}\right) \pmod{p}.$$

$$\left(\frac{ab}{p}\right) \equiv (ab)^{\frac{p-1}{2}} \pmod{p}.$$

$$\therefore \left(\frac{a}{p}\right)\left(\frac{b}{p}\right) \equiv \left(\frac{ab}{p}\right) \pmod{p}.$$

However, $\left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$ & $\left(\frac{ab}{p}\right)$ are either 1 or -1,

and p odd prime implies $p > 2$. So if they are congruent mod p , they must be equal. □

Example: ① $\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \pmod{p}.$

$$\therefore \left(\frac{-1}{p}\right) = \begin{cases} +1 & \text{when } p \equiv 1(4) \\ -1 & \text{when } p \equiv 3(4) \end{cases}.$$

$$\begin{aligned} \text{② } \left(\frac{2}{p}\right) &= (-1)^{\frac{p^2-1}{8}} \\ &= \begin{cases} +1 & \text{when } p \equiv \pm 1(8) \\ -1 & \text{when } p \equiv \pm 3(8) \end{cases} \end{aligned}$$

Theorem: (Quadratic Reciprocity Law).

If p, q are distinct odd primes, then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)}$$

Example:

$$\therefore \left(\frac{53}{2017}\right)\left(\frac{2017}{53}\right) = (-1)^{\left(\frac{53-1}{2}\right)\left(\frac{2017-1}{2}\right)}$$

$$\left(\frac{53}{2017}\right)\left(\frac{2017}{53}\right) = 1$$

$$\therefore \left(\frac{53}{2017}\right) = \left(\frac{2017}{53}\right) \\ = \left(\frac{3}{53}\right)$$

$$\therefore \left(\frac{3}{53}\right)\left(\frac{53}{3}\right) = (-1)^{\left(\frac{3-1}{2}\right)\left(\frac{53-1}{2}\right)} \\ = 1$$

$$\therefore \left(\frac{3}{53}\right) = \left(\frac{53}{3}\right) \\ = \left(\frac{2}{3}\right) \\ = -1$$

$$\therefore \left(\frac{53}{2017}\right) = -1.$$

Exercise:

$$\text{i) } \left(\frac{20964}{1987}\right) \quad \text{ii) } \left(\frac{741}{9283}\right) \quad \text{iii) } \left(\frac{3}{p}\right) \quad \text{iv) } \left(\frac{5}{p}\right).$$

Applications

Example: Solve $x^2 + 3xy - 2y^2 = 122$, $x, y \in \mathbb{Z}$.

Solution: The equation is equivalent to $4x^2 + 12xy - 8y^2 = 488$.

Complete square, $(2x + 3y)^2 - 17y^2 = 488$.

Consider mod 17, $(2x + 3y)^2 \equiv 12 \pmod{17}$.

$$\therefore \left(\frac{12}{17}\right) = \left(\frac{4}{17}\right)\left(\frac{3}{17}\right)$$

$$= \left(\frac{3}{17}\right)$$

$$= -1, \text{ as } 17 \equiv 5 \pmod{12}.$$

$\therefore 12$ is a g.u.r. mod 17, no solution.

Example: Show that there are infinitely many primes of the form $4k+1$.

Solution: Assume the contrary. Let $p_1, \dots, p_k$ be all the primes of the form $4k+1$. Then consider $P = (2p_1 \dots p_k)^2 + 1$.

~~#~~ Consider any prime factor p of P .

$$\text{Then } (2p_1 \dots p_k)^2 + 1 \equiv 0 \pmod{p}.$$

$$(2p_1 \dots p_k)^2 \equiv -1 \pmod{p}.$$

$$\therefore \left(\frac{-1}{p}\right) = 1, \quad p \equiv 1 \pmod{4}.$$

However, $p_i \nmid P$ for all $i=1, \dots, k$. ~~#~~

$\therefore$ There are infinitely many primes of the form $4k+1$.

□

Example: Find all integers a, b s.t. $2b^2 + 3 \mid a^2 - 2$.

* Idea:

- After trying for a while, we want to show that it has no solution.
- $a^2 - 2$ is divisible by something implies $a^2 \equiv 2 \pmod{p}$, some prime factor of $2b^2 + 3$.
- For the congruence equation to have no solution, $p \equiv \pm 3 \pmod{8}$. We know that having one such prime is sufficient to justify our claim.
- Since $2b^2 + 3 \equiv \pm 3 \pmod{8}$, it must have a prime factor $\equiv \pm 3 \pmod{8}$.

Proof:

$\therefore 2b^2 + 3 \equiv \pm 3 \pmod{8}$, it must contain a prime factor p s.t. $p \equiv \pm 3 \pmod{8}$.

Since $2b^2 + 3 \mid a^2 - 2$, in particular we have $p \mid a^2 - 2$.

$\therefore a^2 \equiv 2 \pmod{p}$. However, since $\left(\frac{2}{p}\right) = -1$,

$\therefore$ there is no solution for a, b . □

Problem 1 (IMO 1996 P4).

$$\begin{aligned} \text{Let } 15a + 16b &= r^2 \\ 16a - 15b &= s^2, \quad \text{some } r, s \in \mathbb{N}. \end{aligned}$$

then ~~16a~~ $15r^2 + 16s^2 = 481a$

$481 = 13 \times 37$, so consider mod 13 & mod 37.

$$\begin{aligned} \left(\frac{16s^2}{13}\right) &= \left(\frac{-15r^2}{13}\right) & \text{and} & \quad \left(\frac{16s^2}{37}\right) = \left(\frac{-15s^2}{37}\right) \\ &= \left(\frac{r^2}{13}\right)\left(\frac{-15}{13}\right) & & \quad = \left(\frac{r^2}{37}\right)\left(\frac{-15}{37}\right). \end{aligned}$$

$$\begin{aligned} \therefore \left(\frac{-15}{13}\right) &= -1, \quad \left(\frac{-15}{37}\right) = \left(\frac{15}{37}\right) \\ &= \frac{\left(\frac{37-1}{2}\right)\left(\frac{15-1}{2}\right)}{(-1)} \left(\frac{37}{15}\right) \\ &= \left(\frac{3}{37}\right)\left(\frac{5}{37}\right) \\ &= -1 \end{aligned}$$

$$\therefore \left(\frac{16s^2}{13}\right) = -\left(\frac{r^2}{13}\right) \quad \& \quad \left(\frac{16s^2}{37}\right) = -\left(\frac{r^2}{37}\right)$$

$$\therefore \left(\frac{r^2}{13}\right) = \left(\frac{r^2}{37}\right) = 0, \quad 13|r, \quad 37|r.$$

$$\begin{aligned} \therefore 16s^2 &\equiv 481a - 15r^2 \\ &\equiv 0 \pmod{481}, \quad s \equiv 0 \pmod{481}. \end{aligned}$$

minimum of r^2, s^2 is are both being 481^2 .

This is can be attained when $a = 481 \times 31$,
 $b = 481$.

Problem 2

Since $3 \mid 5+1$, by lifting the Exponent,

$$\begin{aligned} \cancel{v_p} v_3(5^n+1) &= v_3(5+1) + v_3(n) \\ &= 1 + v_3(n). \end{aligned}$$

$$\therefore \text{For } \cancel{5^n+1} \quad 3^n \mid 5^n+1, \quad v_3(5^n+1) \geq n.$$

$$\therefore 1 + v_3(n) \geq n$$

$$v_3(n) \geq n-1.$$

$$3^{n-1} \mid n.$$

When $n=1$, $3^{n-1} = n$, $n=1$ is indeed a solution.

$$\text{When } n > 1, \quad \frac{3^{(n+1)-1}}{3^{n-1}} > \frac{n+1}{n}.$$

$\therefore 3^{n-1} > n$ ~~by induction~~ for all $n > 1$ by induction.

$$\therefore n=1.$$

Problem 3 (IMO 1990 P3) if $n=1$, $1^2 \mid 2^1 + 1$. ✓

Clearly, n odd. ← Consider ^{if $n > 1$,} smallest prime factor p .

$$\text{Then } 2^n + 1 \equiv 0(p) \Leftrightarrow 2^n \equiv -1(p), \\ 2^{2n} \equiv 1(p).$$

Let d be the order of $2 \bmod p$.

Then $d \mid 2n$ & $d \mid p-1$, by F.O.T.

However, since $p \mid n$, $(2n, p-1) = 2$. Thus $d=2$ or 1 .

$$\therefore 2 \not\equiv 1(p), \therefore d=2, \text{ i.e. } 2^2 \equiv 1(p), p=3.$$

Let $v_3(n) = a$. By Lifting the Exponent,

$$v_3(2^n + 1) = v_3(2 + 1) + v_3(n) \\ = 1 + a.$$

$$\therefore n^2 \mid 2^n + 1, \therefore v_3(2^n + 1) \geq 2a.$$

$$1 + a \geq 2a$$

$$a \leq 1.$$

$$\therefore \cancel{a=1} + \cancel{a=1} \text{ ~~when } a=1~~, \\ \therefore a=1$$

$n=3$ is indeed a solution. Otherwise, consider the second smallest prime factor q . Similarly,

letting $f = \text{order of } 2 \bmod q$, $f \mid 2n$ & $f \mid q-1$.

Now $(2n, q-1)$ can be 2 or 6 . In both cases,

$$2^6 \equiv 1(q). \text{ This gives } q=7. \text{ However, } 7 \nmid 2^n + 1.$$

$$\therefore n=3.$$

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